MIL-C-43242C
6 May 1980
SUPERSEDING
MIL-C-43242B
27 September 1976

MILITARY SPECIFICATION

CAKE, RAISIN AND CAKE, RAISIN-NUT; CANNED, BAKED OR STEAMED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers canned, baked or steamed raisin cake and raisin-nut cake, for use by the Armed Forces as a component of operational rations.

1.2 Classification. The product shall be of the following types and styles, as specified (see 6.1).

Type I - Raisin
Type II - Raisin-nut

Style 1 - Baked
Style 2 - Steamed

2. APPLICABLE DOCUMENTS

* 2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Command, Natick, MA 01760, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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FSC 8920

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SPECIFICATION

MILITARY

MIL-C-43991 - Canned Baked, Canned Steamed, Subsistence Items, Packaging Of

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-668 - Sanitary Standards for Food Plants
MIL-STD-1105 - Sanitary Standards for Bakeries

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

LAWS AND REGULATIONS

US DEPARTMENT OF HEALTH, EDUCATION AND WELFARE

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder

(Application for copies should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

US DEPARTMENT OF AGRICULTURE

US Standards for Grades of Shelled Walnuts
US Standards for Grades of Shelled Almonds
US Standards for Grades of Shelled Pecans
US Standards for Grades of Raisins
US Standards for Grades of Nonfat Dry Milk (spray dried)
US Regulations Governing the Grading and Inspection of Egg Products

(Application for copies should be addressed to the Information Division, Agricultural Marketing Service, US Department of Agriculture, Washington, DC 20250.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

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AMERICAN OIL CHEMISTS SOCIETY

Official and Tentative Methods of the American Oil Chemists Society

(Application for copies should be addressed to the American Oil Chemists Society, 508 South Sixth Street, Champaign, IL 61820.)

NATIONAL RESEARCH COUNCIL

Food Chemicals Codex

(Application for copies should be addressed to the Office of Director, National Academy of Science, National Research Council, 2101 Constitution Avenue, N.W., Washington, DC 20418.)

ASSOCIATION OF OFFICIAL ANALYTICAL CHEMISTS

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, 1111 N. Nineteenth St., Suite 210, Arlington, VA 22209.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

* 3.1 First article. When specified (see 6.1), the contractor shall furnish sample units for first article inspection and approval (see 4.4 and 6.3).

3.2 Materials. All materials shall be clean, sound, wholesome, and free from evidence of insect infestation, foreign materials, off-flavors, or off-odors.

3.2.1 Sugar. Sugar shall be white, refined, granulated cane or beet sugar, or a combination of both.

3.2.2 Salt. Salt shall be white, refined, sodium chloride. Iodized salt shall not be used.

* 3.2.3 Enriched flour. Enriched flour shall be wheat flour of a type which will result in the required end product. Enrichment shall be in accordance with the requirements of the FDA and may be accomplished as a part of the milling process or by adding equivalent enrichment to each batch. The flour may be bleached as necessary to provide the required product.

* 3.2.4 Shortening. The shortening shall be refined hydrogenated cottonseed or peanut oil, or a combination of both, and shall have a stability of not less than 100 hours when tested prior to the addition of the antioxidant by the active oxygen method (AOM) (see 4.5.1.2). The shortening shall contain an antioxidant in the form of TBHQ at the maximum allowable level.

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3.2.5 Chemical leavening. Chemical leavening shall be powdered, free from lumps, and of a type which will produce the required product.

3.2.6 Nuts. The nuts shall be No. 1 grade of either pecan, almond or walnut pieces. The nuts shall be nominal 1/8 inch (3.2 mm) to 3/8 inch (9.5 mm). The walnut pieces shall be treated with a 1:1 mixture of BHA and BHT dissolved in vegetable oil. The level of BHA and BHT shall not exceed 0.02 percent, based on the amount of oil contained in the walnut pieces (approximate analysis 64 percent). The nuts shall be of the latest crop.

* 3.2.7 Eggs, whole, fresh, frozen or dried. The egg products shall be whole, fresh, frozen or dried, or a combination thereof. Egg products shall have been produced and handled in compliance with requirements set forth in the USDA publication entitled "Regulations Governing the Inspection of Eggs and Egg Products." They shall be prepared under continuous inspection of the USDA and shall bear the inspection shield mark. If fresh eggs are used, temperatures shall be maintained at or below 40°F (4°C) prior to use; when thawed frozen eggs are used the temperature and time requirements of the above regulations shall be met. When dried eggs are used in the formulation, an equivalency ratio of 1-part by weight of dried eggs to 3-parts by weight of water shall be maintained.

3.2.8 Milk, nonfat, dry. Nonfat, dry milk shall meet requirements for extra grade as defined in the US Standards for Grades of Nonfat Dry Milk (spray dried).

3.2.9 Vanilla flavoring. Vanilla flavoring may be natural or artificial and of a type and strength that will produce a mild but organoleptically detectable vanilla flavor.

3.2.10 Raisins. Raisins shall be select soda dipped or bleached Thompson seedless raisins, grade B or better as defined in US Standards for Grades of Raisins.

3.3 Formula and preparation.

* 3.3.1 Formula. The product shall be prepared in accordance with the following formula:

Materials	Parts by weight
Sugar	21.5
Flour	21.5
Nonfat dry milk	1.0
Chemical leavening 3/	As required
Salt (added)	0.5
Shortening	15.0
Eggs (fresh, frozen, or dried (on a liquid basis) 2/	15.0
Vanilla 3/	As required
Raisins, whole	25.5
Water 3/	Variable

1/ To produce raisin-nut cake add 12 parts by weight of nuts to each 100 parts by weight of the above formula.

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2/ When dried whole eggs are used the reconstitution ratio shall be one part dried eggs per three parts water.

3/ Weight in addition to total formula weight.

3.3.2 Preparation.

* 3.3.2.1 Pre-treatment of raisins. Raisins shall be dusted with flour and blended thoroughly so that each raisin is coated with flour sufficiently to prevent clumping of raisins within the batter.

3.3.2.2 Blending. The batter shall be blended (raisins or raisins and nuts, as applicable) in such a manner as to yield a finished product complying with 3.4.

* 3.3.2.3 Depositing of batter. A predetermined quantity of batter, sufficient to insure a baked or steamed end item meeting the pertinent net weight requirements of 5.1 shall be deposited in accordance with the requirements of MIL-C-43991. Alternatively in lieu of paper disc and liner, the can shall have an interior enamel, possessing adequate product release characteristics, so that not more than 5.0 percent by weight of the product adheres to the interior of the can. The lids of the cans shall be clinched on loosely to allow for the escape of moisture and gases evolved during processing.

* 3.3.2.4 Processing style 1 (baked). The filled cans with loosely clinched lids shall be placed in an oven and baked at not more than 375°F (191°C) for as much time as is required for the finished product to meet the end item requirements of 3.4 and to a temperature at the center of the product of not less than 195°F (91°C) at the time of sealing. The cans shall be removed from the oven and hermetically sealed under atmospheric conditions. The cans shall have retained sufficient oven heat at the time of sealing so that in cooling to room temperature, the internal vacuum shall be not less than 25 inches of mercury gauge (not more than 16.7 kPa absolute pressure). Within 1 hour after the can is hermetically sealed, the center of the product shall be cooled to a temperature of less than 100°F (38°C) or less than 10°F (6°C) above the ambient temperature, whichever is higher. The entire processing procedure, including can cooling shall be performed in such a manner that the resulting product shall meet the end item requirements of 3.4.

* 3.3.2.5 Processing style 2 (steamed). The filled cans with loosely clinched lids shall be processed in saturated steam at atmospheric pressure until adequately processed in accordance with 3.4. The temperature at the center of the product shall be not less than 195°F (91°C) at the time of sealing. Immediately after being discharged from the steamer, the cans shall be hermetically sealed. Sealing of the cans shall be so timed that after cooling to room temperature, the internal vacuum shall be not less than 10 inches of mercury gauge (not more than 67.5 kPa absolute pressure). Within 1 hour after the can is hermetically sealed, the center of the product shall be cooled to a temperature of less than 100°F (38°C) or less than 10°F (6°C) above the ambient temperature, whichever is higher.

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* 3.4 Finished product. The finished product shall conform to the inside contour of the can or can liner. Neither the product nor the can liner shall interfere with the compound in the double seam in such a way as to disrupt the continuity of the seal. The distance from the edge of the can to where the cake touches the top lid during heat processing shall not exceed 3/4 inch (19 mm). The processing time shall be adjusted so that the batter portion of the cake is heated uniformly throughout to produce a finished product having no raw, stringy or ungelatinized portions. When the cooled product is bisected vertically and horizontally with a sharp knife, it shall not crumble nor show any compression streaks, gummy centers, soggy areas nor be excessively dry or over processed and shall display an even grain structure throughout. There shall be no scorched or burnt spots. The raisins shall be whole raisins. The raisins or nuts (as applicable) shall be evenly blended and distributed throughout the sliced surface. The flavor shall be typical of the flavor combination or raisins and pound cake and for raisin-nut cake typical of a combination of nuts, raisins and pound cake. The product shall have no off flavors or off odors. The water activity of the cakes shall not be greater than 0.90.

3.5 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

3.6 Workmanship. The product shall be prepared, processed and packaged under modern sanitary conditions and in accordance with good commercial practice. The product shall be processed in establishments meeting the requirements of MIL-STD-1105 or if produced in food establishments other than bakeries shall meet the requirements of MIL-STD-668.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Plant qualification (see 4.3).
b. First article inspection (see 4.4).
c. Quality conformance inspection (see 4.5).

4.3 Plant qualification. Product shall be rejected if produced in plants not meeting the requirements of MIL-STD-1105, or if produced in food establishments other than bakeries, not meeting the requirements of MIL-STD-668.

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* 4.4 First article inspection. When preproduction samples of the product which the contractor intends to supply shall be inspected to assure compliance with 3.4.

4.5 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.5.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.5.1.1 Examination of ingredients. Examination shall be in accordance with 4.5.1.1.1 and 4.5.1.1.2. Nonconformance with identity or condition requirements shall indicate an unacceptable ingredient and use of such shall be cause for rejection of the involved finished product.

* 4.5.1.1.1 Identity. Examination shall be made of label, invoices or other valid documents to determine conformance to identity requirements (kinds, grade standards or other standards, strength, size, source of inspection, additives, etc., as pertinent to the ingredient).

* 4.5.1.1.2 Condition. In addition, each ingredient shall be examined organoleptically as necessary to determine conformance to condition requirements of 3.2. The sample unit for examination shall be approximately one pound (0.45 kg) of each ingredient (except for vanilla flavoring which shall be four fluid ounces (118 ml) and chemical leavening which shall be 1/2 pound) (0.23 kg) taken from each of five primary containers. The sample size shall be five. In the event a lot consists of less than five containers a sample unit shall be extracted from each container in the lot.

* 4.5.1.1.3 Examination of eggs, whole, fresh, frozen or dried. Egg products shall comply with the requirements of 3.2.7. Use of noncomplying egg products shall be cause for rejection of the involved quantity of the finished product.

* 4.5.1.2 Testing of shortening. Testing of shortening as specified in 3.2.4 shall be as shown in table I. Results shall apply to the lot average. Nonconformance to the test requirement shall be cause for rejection of the lot.

TABLE I. Testing of component

Component	Sample unit	Lot size expressed in	Inspection level	Characteristics	Results reported	Test ref and rqmt para
Shortening	1 lb composite 1/ (0.45 kg composite)	Pounds (kg)	S-2	Stability (AOM)	Pass or fail 2/	3.2.4 4.6.1

1/ The composite shall be derived from the number of primary containers as indicated by the inspection level.

2/ If failure, report results to nearest hour.

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* 4.5.2 In-process inspection. Examination shall be performed to determine conformance to formulation and processing requirements of 3.3. Records of formulation and processing temperatures shall be maintained. In addition, the process examination of the cooked product as specified in the quality assurance provisions of MIL-C-43991 shall be performed. Nonconformance shall be cause for rejection of the involved product.

* 4.5.3 End item inspection. Examination shall be in accordance with table III thru V. The sampling plans for these tables shall be in accordance with table II. The acceptable quality levels (AQLs) shall be expressed as percent defective for table III and defects per hundred units for table IV and table V.

TABLE II. Criteria for end item sampling

Table	Inspection level	Sample unit	Lot size expressed in	AQLs Major	AQLs Minor
III	S-3	1 filled and sealed can	Cans	-	2.5
IV	S-2	1 filled and sealed can	Cans	1.5	4.0
V	S-2	1 filled and sealed can	Cans	-	4.0

TABLE III. Examination for net weight

Category	Defect
Minor	
201	Less than 4 ounces (113.4 g) (300 x 200 can). 1/

1/ Report results to nearest 1/16 ounce (or 1g). Lot shall be rejected if sample data indicate a lot average net weight of less than 4-1/4 ounces (120.5 g).

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TABLE IV. Examination for product characteristics 1/ 2/

Category		Defect
Major A	Minor	
101		Scorched or burnt spots.
102		Raw, stringy, or ungelatinized portions.
103		When bisected vertically and horizontally with a sharp knife: a. Product crumbles, or b. Shows compression streaks, or c. Is excessively dry, or d. Has soggy or gummy areas e. Grain structure not even throughout.
104		Flavor not typical of the flavor combination(s) and pound cake, as applicable.
	201	Raisins not identifiable as whole raisins (cut or macerated).
	202	Raisins and pieces of nuts (when applicable) missing.
	203	Raisins and pieces of nuts (when applicable) not uniformly distributed.

1/ Presence of foreign material such as dirt, paint, wood, metal, insect and insect parts shall be cause for rejection of the lot.

2/ Presence of rancid, stale or foreign or objectionable flavors or odors shall be cause for rejection of the lot.

TABLE V. Examination for fill of container

Category	Defect
Minor	
201	Distance from edge of can to where cake touches the top lid exceeds 3/4 inch (19 mm).
202	Lack of continuity in compound seal. 1/

1/ Examine only in cans in which the product completely fills the can, touching the top lid at a point closer than 1/4 inch from edge. Examination of can seams and tear down procedure shall be made in accordance with the Code of Federal Regulations, Title 21, Parts 100-199, Subpart D, Part 113.60.

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* 4.5.4 Packaging inspection. The packaging inspection for the section 5 requirements shall be in accordance with the applicable quality assurance provisions of MIL-C-43991.

* 4.5.5 Testing of finished product. The finished product shall be tested for water activity in accordance with 4.6. Testing shall be on a unit basis. The sample unit for testing shall be one filled and closed can. Lot size shall be expressed in cans. The sample size shall be the number of sample units indicated by inspection level S-2. The AQL expressed as percent defective shall be 1.0. Results shall be reported to the nearest 0.01 unit. A test result not in conformance shall be scored as a defect.

* 4.6 Methods of inspection.

* 4.6.1 Chemical analyses. Chemical analyses shall be made in accordance with the following methods from the publications referenced below. Unless otherwise indicated, references are to Official Methods of Analysis of the Association of Official Analytical Chemists:

Test	Source	Method
Shortening stability (AOM)	Section C, Chapter: Commercial Fats and Oils 1/	Fat stability Cd 12-57
Finished product:		
Water activity	Chapter: Vegetable Products, Processed Section: Canned Products	Water activity

1/ Official and Tentative Methods of the American Oil Chemists Society.

5. PACKAGING

5.1 Preservation, packing, unit loading, labeling and marking.

5.1.1 Levels A and C. A unit pack net weight of four and one-quarter ounces (120.5 g) of product shall be preserved, packed, unit loaded, labeled and marked in accordance with the applicable requirements of MIL-C-43991 (see 6.1). The net weight of the product in size 300 x 200 cans shall be not less than 4-1/4 ounces (120.5 g). A tolerance of minus 1/4 ounce (7 g) is permitted provided that the average net weight of cans in any lot is not less than 4-1/4 ounces (120.5 g).

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type and style of product required (see 1.2).
(c) When a first article is required (see 3.1, 4.4 and 6.3).
(d) Level of preservation required (see 5.1.1).

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* 6.2 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

* 6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of ASPR 7-104.55. The first article shall be a preproduction sample and shall consist of ten cans of the product. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the first article.

* 6.4 Metric equivalents. Metric equivalents, indicated in parentheses throughout this document, are based on practices, conversion factors, and symbols specified in ASTM E 380 Standard for Metric Practice, and are for information only. In each instance, the value stated in US customary units shall be controlling.

Custodians:

Army - GL
Navy - SA
Air Force - 50

Preparing activity:

Army - GL

Project No. 8920-0412

Review activities:

Army - MD
Navy - MC, MS
DP - SS

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